Answer:

Our nodal equations are:

$$5 = \frac{v_1}{1} + \frac{v_1 - v_2}{5} \quad [1]$$

$$-4 = \frac{v_2}{2} + \frac{v_2 - v_1}{5} \quad [2]$$

Solving, $v_1 = 3.375$ V and $v_2 = -4.75$ V.

Hence,

$$i = (v_1 - v_2)/5 = 1.625 \text{ A}$$

Answer:

(a)

Define CW mesh currents i_1, i_2 and i_3 such that $i_3 - i_2 = i$.

Our mesh equations then are:

$$0 = -2 + 8i_1 - 4i_2 - i_3$$

$$0 = 5i_2 - 4i_1$$

$$0 = 5i_3 - i_1$$

Solving, $i_1 = 434.8$ mA, $i_2 = 347.8$ mA, and $i_3 = 86.96$ mA

Then, $i = i_3 - i_2 = -260.8$ mA

(b) According to voltage division $v_1=v_2=2/4.6*0.8=347$ mV

(c) It make sense because the resistance of the wire is 0, even if the voltage difference between the two ends is 0, it does not conflict with the current flowing through it.